plmc

A minimal PL/M compiler for the Motorola 6809 – user manual

plm09 project

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1 Introduction

`plmc` compiles programs written in a small subset of PL/M-80, Intel's systems programming language of the late 1970s, into assembly source for the Motorola 6809. Its output is meant for **lwasm**, the assembler in the lwtools package (version 4.24 or later). lwasm then produces whatever binary format the target needs: S-records, Disk BASIC `.BIN` files, raw images and so on.

The compiler is not tied to any particular machine. Everything that depends on the hardware lives in a small **hooks file** that lwasm includes, chosen with an include path. Three hooks files come with plmc:

Hooks	Target
`runtime/sim/`	`sim6809`, the headless simulator used by the test suite
`runtime/coco/`	the Tandy Color Computer 1, 2 and 3 under Color BASIC
`runtime/example/`	a template for a board with an MC6850 ACIA serial port

Main features

- BYTE (8-bit) and ADDRESS (16-bit) unsigned variables, with PL/M-80's rules for mixing the two types.
- All PL/M arithmetic, logical and relational operators, with PL/M precedence.
- `IF`/`THEN`/`ELSE`, `DO` blocks with their own declarations, `DO WHILE` loops, `HALT`.
- Procedures: typed and untyped, with parameters, nested inside each other, and `EXTERNAL` procedures written in assembly.
- Error messages that give the file, line and column. Every unsupported PL/M feature is rejected with a clear message rather than misread.
- Readable output: each source statement appears as a comment above the code generated for it.

2 Building and quick start

`plmc` is plain C11. From the top of the plm09 tree:

```
make            # builds build/plmc and build/sim6809
make test       # runs every test suite
```

A first program:

```
SUM: DO;
PUTDEC: PROCEDURE (N) EXTERNAL; DECLARE N ADDRESS; END PUTDEC;
DECLARE I BYTE, TOTAL ADDRESS;
TOTAL = 0;
I = 1;
DO WHILE I <= 10;
    TOTAL = TOTAL + I;
    I = I + 1;
END;
CALL PUTDEC(TOTAL);
END SUM;
```

Compile it, assemble it, and run it in the simulator:

```
$ build/plmc -o sum.asm sum.plm
$ lwasm -9 -f srec -I runtime -I runtime/sim -o sum.s19 sum.asm
$ build/sim6809 sum.s19
55
```

`make run F=sum.plm` does all three steps. Always pass `-9` to lwasm (or rely on the `PRAGMA 6809` that plmc emits). By default lwasm accepts the Hitachi 6309's extra instructions, which a 6809 doesn't have.

3 Command-line usage

```
plmc [options] file.plm
```

Option	Meaning
`-o FILE`	Write the assembly to *FILE*. The default is standard output.
`--org ADDR`	Address where the program is loaded and starts. The default is `$0100`.
`--vectors`	Also emit `ORG $FFFE` / `FDB RT_START`, setting the 6809 reset vector, for programs that run from ROM or straight from reset.
`--tokens`	Print the token stream and stop. See Inspecting the compiler's view.
`--ast`	Print the syntax tree and stop.
`-h`, `--help`	Print a usage summary and exit.

The `--org` address can be written as decimal (`8192`), `0x2000`, `$2000` (quote it for the shell) or PL/M style `2000H`. It must fit in 16 bits.

Exit status

Status	Meaning
0	Success. Any warnings are still printed.

Status	Meaning
1	Compilation error, or a file couldn't be read or written. If `-o` was given and writing failed, the partial output file is removed.
2	Bad usage, for example no input file or an unknown option.

4 The language

plmc accepts a **subset** of PL/M-80. Anything outside the subset is rejected with an error (see Unsupported features), so a program that compiles means what PL/M-80 says it means.

4.1 Lexical rules

- **Case:** case doesn't matter. `Total`, `TOTAL` and `total` are the same name, and keywords may be written in any case.
- **The `$` character:** ignored inside names and numbers, so you can use it as a separator: `MAX$COUNT`, `1$000`, `0FF$FFH`.
- **Names:** start with a letter and continue with letters, digits and `$`. The underscore is **not** allowed (as in PL/M-80). This is what keeps compiler-generated labels from ever colliding with your names.
- **Comments:** `/* ... */`. They may span several lines but don't nest.
- **Numbers:** must start with a digit and fit in 16 bits (0 to 65535). An optional suffix gives the base:

Suffix	Base	Example	Value
none or `D`	decimal	`255`, `255D`	255
`H`	hexadecimal	`0FFH`	255
`O` or `Q`	octal	`377Q`	255
`B`	binary	`1111$1111B`	255

  A hexadecimal number has to start with a digit, so write `0FFH`, not `FFH` (`FFH` is a name).

- **Strings:** a one-character string such as `'A'` is a BYTE constant (65). A two-character string such as `'AB'` is an ADDRESS constant (`'A'`× 256 + `'B'`). Write a quote character as `''`, as in `''''`. Longer strings are not supported.
- **Reserved words:** all PL/M-80 reserved words, including ones for features plmc doesn't implement, so they can't be used as names: `ADDRESS AND BASED BY BYTE CALL CASE DATA DECLARE DISABLE DO ELSE ENABLE END EOF EXTERNAL GO GOTO HALT IF INITIAL INTERRUPT LABEL LITERALLY MINUS MOD NOT OR PLUS PROCEDURE PUBLIC REENTRANT RETURN STRUCTURE THEN TO WHILE XOR`.

4.2 Program structure

A program is one labeled block:

```
NAME: DO;
    declarations
    statements
END NAME;
```

The name after `END` is optional, but if present it must match. The block may optionally be followed by the keyword `EOF`. Within any block, **all declarations must come before the first statement.**

4.3 Declarations

```
DECLARE COUNT BYTE;
DECLARE (X, Y, Z) ADDRESS;
DECLARE FLAG BYTE, LIMIT ADDRESS, (LO, HI) BYTE;
```

- **Types:** `BYTE` is an unsigned 8-bit value (0 to 255) and `ADDRESS` an unsigned 16-bit value (0 to 65535).
- **Order and scope:** names must be declared before they are used. A name declared in an inner block or procedure hides the same name from an outer one. Declaring a name twice in the same block is an error.
- **Initial values:** variables have no initial value. They start with whatever is in memory, which is `$A5` in the simulator so that the mistake shows up.

4.4 Procedures

```
NAME: PROCEDURE [(PARAM, ...)] [BYTE | ADDRESS] [EXTERNAL];
    declarations
    statements
END [NAME];
```

- **Parameters:** each one must be declared, with its type, in the procedure's own declarations.
- **Typed and untyped:**
  - A typed procedure (`BYTE` or `ADDRESS`) returns a value with `RETURN expression;` and is used inside expressions.
  - An untyped procedure is invoked with `CALL` and may use a bare `RETURN;`.
  - Using one kind where the other is expected is an error.
- **No arguments:** a typed procedure with no parameters is called just by writing its name: `X = NEXTCHAR;`.
- **Nesting:** procedures may be nested. An inner procedure can use the parameters and variables of the procedures around it, and of the program.
- **Argument conversion:** arguments are converted to the parameter's declared type, and `RETURN` values to the procedure's type. See Types and conversions.
- **Reaching `END`:** an untyped procedure just returns. A typed procedure that might reach its `END` without a `RETURN` gets a warning, and the value it returns is then undefined.
- **No recursion:** as in classic PL/M-80, parameters and local variables live at fixed memory locations, not on the stack. A procedure must not call itself, directly or through other procedures. plmc reports both cases:

  ```
  error: procedure A is recursive (A -> B -> A); recursion needs
         REENTRANT, which is not supported
  ```

4.4.1 EXTERNAL procedures

An `EXTERNAL` procedure is written in assembly language, usually in the hooks file. Its PL/M declaration gives only the parameters and their types:

```
PUTCH: PROCEDURE (C) EXTERNAL;
    DECLARE C BYTE;
END PUTCH;
```

The body may contain only the parameter declarations. In the assembly the procedure is called by its exact PL/M name, so it must not be a 6809 register name (`A`, `B`, `D`, `X`, `Y`, `U`, `S`, `PC`, `CC`, `DP`, `PCR`). The calling convention is described under Hooks.

4.5 Statements

Statement	Form
Assignment	`X = expression;`
Multiple assignment	`X, Y, Z = expression;` (evaluated once, converted for each target)
Call	`CALL NAME;` or `CALL NAME(arg, ...);` (untyped procedures only)
Conditional	`IF condition THEN statement; [ELSE statement;]`
Block	`[LABEL:] DO; declarations statements END [LABEL];`
Loop	`[LABEL:] DO WHILE condition; statements END [LABEL];`
Return	`RETURN;` or `RETURN expression;` (inside procedures only)
Halt	`HALT;` stops the program through the hooks' `PLM_HALT`
Empty	`;`

- **ELSE:** an `ELSE` belongs to the nearest `IF` that doesn't already have one. To put several statements under `THEN` or `ELSE`, wrap them in `DO; ... END;`.
- **Labels:** only `DO` blocks may be labeled, and the label exists only so that `END` can be checked against it.
- **Loop blocks:** a `DO WHILE` block may not contain declarations.

4.6 Expressions

Operators, from highest to lowest precedence:

Precedence	Operators	Notes
1 (highest)	`( )`	grouping
2	unary `-`, unary `+`	negation (two's complement)
3	`* / MOD`	multiply, divide, remainder
4	`+ -`	add, subtract
5	`= <> < > <= >=`	comparisons
6	`NOT`	bitwise complement

Precedence	Operators	Notes
7	`AND`	bitwise and
8 (lowest)	`OR XOR`	bitwise or, exclusive or

- **Grouping:** operators of equal precedence group left to right, so `A - B - C` is `(A - B) - C`. Comparisons do too: `A < B = C` compares the result of `A < B` with `C`.
- **Low precedence of `NOT`:** `NOT` binds more loosely than the comparisons, so `NOT A < B` means `NOT (A < B)`.
- **Evaluation order:** the order in which operands are evaluated is not defined, as in PL/M. Avoid expressions where a procedure call changes a variable used elsewhere in the same expression.

4.7 Types and conversions

The type of every expression is worked out at compile time:

Expression	Result type
constant 0–255	BYTE
constant 256–65535, or a two-character string	ADDRESS
variable	its declared type
typed procedure call	the procedure's type
`a + b`, `a - b`, `a MOD b`, `a AND b`, `a OR b`, `a XOR b`	BYTE if *both* operands are BYTE, otherwise ADDRESS
`a * b`, `a / b`	always ADDRESS
`a = b`, `a < b`, …	BYTE: `0FFH` for true, `0` for false
`-a`, `NOT a`	the type of `a`

Rules that follow from this table:

- **BYTE arithmetic wraps at 8 bits, before any widening.** If `B` is a BYTE holding 200, `B + 100` is 44, *even when the result is assigned to an ADDRESS*. When you need the full result, make one operand an ADDRESS, for example by copying `B` into an ADDRESS variable first: `W = B; W = W + 100;`.
- **A constant above 255 makes the operation 16-bit.** `B + 300` is ADDRESS arithmetic.
- **Multiplication never wraps at 8 bits.** `200 * 200` is 40000.
- **Unsigned only.** All arithmetic and comparisons are unsigned. `0FFFFH > 1` is true.
- **Assignment converts:** an ADDRESS stored in a BYTE keeps its low byte, and a BYTE stored in an ADDRESS is zero-extended. The same applies to arguments and `RETURN` values.
- **Division by zero** gives a quotient of `0FFFFH` and a remainder equal to the dividend. Programs should not rely on this.

4.8 Conditions

`IF` and `DO WHILE` test **only bit 0** of their condition. A comparison gives `0FFH` or `0`, so this works as expected for comparisons and for `AND`, `OR` and `NOT` applied to them. For other values it means `IF 2` is false and `IF 3` is true. `AND` and `OR` always evaluate both operands; they do not short-circuit.

4.9 Unsupported features

These parts of PL/M-80 are recognized and rejected with an error:

Feature	Error message
arrays, `DECLARE X(10) BYTE`, `X(I) = ...`	`arrays is not supported` / `subscripted assignment (arrays) is not supported`
`BASED` variables, `STRUCTURE`, `LABEL` declarations	`... is not supported`
`LITERALLY` macros	`LITERALLY is not supported`
`INITIAL` and `DATA`	`initialized data is not supported`
the address-of operator `.X`	`the address-of operator '.' is not supported`
iterative `DO I = 1 TO 10` and `DO CASE`	`iterative DO is not supported`, `DO CASE is not supported`
`GOTO`, `GO TO`	`GOTO is not supported`
a label on anything but `DO`	`a label on this statement (only DO blocks may be labeled) is not supported`
`REENTRANT`, `INTERRUPT`, `PUBLIC`	`... is not supported`
`PLUS` and `MINUS` (add and subtract with carry)	`PLUS/MINUS (add/subtract with carry) is not supported`
embedded assignment `:=`	`embedded assignment ':=' is not supported`
`ENABLE`, `DISABLE`	`ENABLE/DISABLE is not supported`
strings of three or more characters	`strings longer than 2 characters are not supported`

Built-in procedures such as `HIGH`, `LOW`, `SHL`, `SHR`, `ROL`, `ROR`, `LENGTH` and `LAST` are not predefined. Using one gives `... is not declared`, unless you write it yourself as a procedure or an `EXTERNAL`.

5 Diagnostics

Errors and warnings go to standard error in the form `file:line:column: error: message`:

```
$ build/plmc e1.plm
e1.plm:3:5: error: Y is not declared
```

Compilation stops at the first error. Warnings don't stop compilation and don't change the exit status. Columns count from 1, and tabs advance to the next multiple of 8 plus 1.

Message	Cause
`expected X, found Y`	syntax error; *X* is what the parser needed at this point
`X is not declared`	name used before (or without) its declaration
`X is already declared at line N`	second declaration in the same scope
`parameter X of P is not declared`	parameter without a `DECLARE` in the procedure
`procedure P expects N arguments, got M`	wrong number of arguments
`X is not a procedure`	`CALL X` or `X(...)` where X is a variable
`typed procedure F must be used in an expression, not CALLed`	`CALL` of a BYTE or ADDRESS procedure
`untyped procedure P cannot be used in an expression`	value taken from an untyped procedure
`cannot assign to procedure P`	procedure name on the left of `=`
`RETURN outside a procedure`	`RETURN` in the main program
`untyped procedure P cannot return a value` / `RETURN in typed procedure F needs a value`	`RETURN` does not match the procedure's type

Message	Cause
`procedure P is recursive (...)`	direct or indirect recursion
`END X does not match ...`	label after `END` differs from the block or procedure
`declarations must come before statements`	`DECLARE` or procedure after the first statement
`declarations are only allowed in procedures and simple DO blocks`	`DECLARE` inside `DO WHILE`
`an EXTERNAL procedure may only declare its parameters` / `... cannot have statements`	body of an `EXTERNAL` procedure
`EXTERNAL name X is also a 6809 register name`	external called `A`, `X`, `PC`, …
`invalid digit`, `invalid number`, `number ... does not fit in 16 bits`	bad numeric constant
`invalid character`, `unterminated comment`, `unterminated string`, `empty string`	lexical errors
warning: `typed procedure F may reach its END without RETURN`	last statement is not `RETURN` or `HALT`

6 Hooks

6.1 How targets are selected

The assembly plmc writes always ends with:

```
INCLUDE "plm_rt.asm"
INCLUDE "plm_hooks.asm"
```

lwasm searches its `-I` directories for these files. `plm_rt.asm` and `plm_crt0.asm` come from `runtime/`. Which `plm_hooks.asm` is used depends on which hooks directory is on the include path, so **the same `.asm` file can be assembled for different machines** without recompiling:

```
lwasm -9 -f srec -I runtime -I runtime/sim  -o prog.s19 prog.asm   # simulator
lwasm -9 -f decb -I runtime -I runtime/coco -o PROG.BIN prog.asm   # CoCo
```

6.2 What a hooks file must define

Symbol	Purpose
`RT_STACKTOP`	Initial stack pointer; the stack grows down from here. The value 0 means "keep the caller's stack".
`PLM_EXIT`	Jumped to when the program ends, with the exit status in B (0 for a normal end). Must not return.
`PLM_HALT`	Jumped to by `HALT`.
each `EXTERNAL`	A label with the procedure's PL/M name in upper case.

The startup code saves the stack pointer it was entered with in `RT_ENTRYS`. A hooks file that sets `RT_STACKTOP` to 0 can therefore return to whatever started the program (BASIC, a monitor or an operating system) with `LDS RT_ENTRYS` / `RTS`. The CoCo hooks do this.

6.3 Calling convention for EXTERNAL procedures

- **Arguments:** each one is pushed as a **16-bit word**, from left to right. BYTE values are zero-extended first.
- **Where to find them:** on entry, the last argument is at `2,S`, just above the return address, the one before it at `4,S`, and so on. The last argument is **also in D**, so most one-argument hooks never need to touch the stack.
- **Cleanup:** the caller removes the arguments after the call (`LEAS 2n,S`).
- **Results:** a BYTE result is returned in **B** and an ADDRESS result in **D**.
- **Registers:** a hook may change A, B, X, Y, U, DP and CC. Only S must be preserved. Keep DP at 0 if the hook calls code that relies on it.
- **Runtime routines:** hooks are assembled after `plm_rt.asm`, so they may call `RT_MUL16` and `RT_DIV16`.

Example: a two-argument hook `POKE(ADDR, VALUE)`:

```
; POKE: PROCEDURE (ADDR, VALUE) EXTERNAL; DECLARE ADDR ADDRESS, VALUE BYTE;
POKE    LDX     4,S         ; first argument: ADDR
        STB     ,X          ; last argument: VALUE, also in D (B = low byte)
        RTS
```

6.4 Hooks provided

All three hooks files implement the same procedures, so a program that uses only these runs unchanged on any of the targets:

```
PUTCH:  PROCEDURE (C) EXTERNAL;    DECLARE C BYTE;    END PUTCH;
PUTDEC: PROCEDURE (N) EXTERNAL;    DECLARE N ADDRESS; END PUTDEC;
GETCH:  PROCEDURE BYTE EXTERNAL;                      END GETCH;
EXIT:   PROCEDURE (CODE) EXTERNAL; DECLARE CODE BYTE; END EXIT;
```

`PUTDEC` prints a number in decimal followed by a newline.

	`runtime/sim`	`runtime/coco`	`runtime/example`
`PUTCH`	simulator I/O port	Color BASIC CHROUT; LF sent as CR	MC6850 ACIA, waits until ready to send
`GETCH`	stdin; 0 at end of input	waits for a key (POLCAT); ENTER gives 13	ACIA, waits for a character
`PUTDEC`	simulator decimal port	via `RT_DIV16` and CHROUT	via `RT_DIV16`, ends with CR LF
`EXIT`, end of program	simulator exit status	back to BASIC (status discarded)	stops in a loop
`HALT`	exit status 0	back to BASIC	stops in a loop
extra		`CLS` clears the screen	
stack	`$FF00`	BASIC's	`$8000`

6.5 Writing hooks for a new machine

Copy `runtime/example/plm_hooks.asm` to a directory of your own, change the I/O routines and `RT_STACKTOP`, and assemble with `-I runtime -I yourdir`. Add any extra routines your programs need as further labels, and declare them in PL/M as `EXTERNAL`. The example file's `ACIA_BASE` and `RT_STACKTOP` are wrapped in `IFNDEF`, so they can also be overridden from the lwasm command line with `-D`.

7 Running on a Color Computer

```
build/plmc --org 3000H -o hello.asm hello.plm
lwasm -9 -f decb -I runtime -I runtime/coco -o HELLO.BIN hello.asm
```

Copy `HELLO.BIN` to a disk or disk image (for example with toolshed: `decb copy -2 -b HELLO.BIN disk.dsk,HELLO.BIN`), then in BASIC:

```
CLEAR 200,&H2FFF
LOADM"HELLO":EXEC
```

- **Load address:** always give `--org`. The default `$0100` would overwrite BASIC's RAM interrupt vectors. `CLEAR 200,&H2FFF` stops BASIC from using memory from `$3000` up.
- **Reset vector:** never use `--vectors` on a CoCo; `$FFFE` is in ROM.
- **Running and returning:** the program runs with DP = 0 on BASIC's own stack, and returns to BASIC when it ends, on `HALT`, or on `EXIT`.
- **Newlines:** Color BASIC ignores the LF character, so `PUTCH(10)` is sent as CR, the CoCo's newline. A program written for the simulator therefore prints the same way on the CoCo.
- **Lower case:** on a CoCo 1 or 2, lower-case letters appear as inverse video.

The test suite checks these hooks in two ways. They run in the simulator against a stand-in for the BASIC ROM. They also run on an emulated CoCo 2 under XRoar: each program is loaded from a disk image with `LOADM` and `EXEC`, and the text screen is compared with what's expected.

8 Generated code

8.1 Layout

```
        PRAGMA  6809
        ORG     $0100                   ; --org
        INCLUDE "plm_crt0.asm"          ; RT_START: set S and DP, call PLM_MAIN
PLM_MAIN                                ; main program statements
        ...
        RTS
P_PROC1                                 ; each procedure
        ...
        INCLUDE "plm_rt.asm"            ; RT_MUL16, RT_DIV16
        INCLUDE "plm_hooks.asm"         ; target-specific hooks
V_X     RMB     1                       ; variables, parameters, locals
        END     RT_START
```

- **Startup:** execution starts at `RT_START`, at the `--org` address. It sets the stack (unless `RT_STACKTOP` is 0) and DP = 0, calls `PLM_MAIN`, and jumps to `PLM_EXIT` with status 0.
- **Variables** come last and are reserved with `RMB`, so they add nothing to S-record or `.BIN` files.
- **Total size:** for a typical small program, the code is a few hundred bytes and the runtime about 70 bytes.

8.2 Example

The program from Building and quick start compiles to (in part):

```
; 6: DO WHILE I <= 10;
L1
        LDB     V_I
        CMPB    #$0A
        LBHI    L2
; 7: TOTAL = TOTAL + I;
        LDB     V_I
        CLRA
        PSHS    D
        LDD     V_TOTAL
        ADDD    ,S++
        STD     V_TOTAL
; 8: I = I + 1;
        LDB     V_I
        ADDB    #$01
        STB     V_I
        LBRA    L1
L2
```

Each statement is preceded by its source line as a comment, which makes the listing and simulator traces easy to follow.

8.3 Conventions

- **Registers:** a BYTE value is computed in B and an ADDRESS value in D. A BYTE is widened with `CLRA` only where a 16-bit value is needed.
- **Binary operators:** the right operand is evaluated first and pushed, then the left one, then they're combined with `op ,S+` or `op ,S++`. When the right operand is a constant or a variable of the right size, it's used directly as the instruction operand (`ADDB V_I`, `CMPB #$0A`).
- **Comparisons:** these use the unsigned branches `BLO`, `BHS`, `BHI`, `BLS`, `BEQ` and `BNE`.
  - In an `IF` or `WHILE`, the code branches straight on the comparison.
  - As a value, a comparison produces `$FF` or `$00` in B.
  - Statement-level control flow uses long branches, so there's no limit on the size of a block.
- **Multiplication and division:** BYTE × BYTE uses the 6809's `MUL`. All other multiplications and all divisions call `RT_MUL16` or `RT_DIV16` (left operand in X, right in D).
- **Procedure calls:** arguments are stored in the procedure's fixed parameter locations and the procedure is called with `JSR`. If an argument itself contains a call, all arguments are evaluated and pushed first, then stored. That makes `F(1, F(2, 3))` work even though `F` has only one set of parameter locations.
- **Label names:** PL/M names can't contain `_`, so these labels never clash with yours:

Label	Meaning
`V_X`	variable `X` of the program
`Vn_X`	variable or parameter `X` of nested scope n
`P_X`, `Pn_X`	procedure `X`
`Lnnn`	compiler-generated label
`RT_*`	runtime routine or data
`PLM_*`	entry point the hooks provide
`X`	`EXTERNAL` procedure `X`

9 Inspecting the compiler's view

`--tokens` prints each token with its line and column. Numbers are shown with their value, which is useful for checking how a constant was read:

```
2:1 IDENT X
2:3 PUNCT =
2:5 NUMBER 255
2:10 PUNCT +
```

`--ast` prints the syntax tree after parsing, with expressions written as prefix lists. It shows exactly how precedence was applied:

```
PROGRAM SUM
  PROCEDURE PUTDEC (N) EXTERNAL
    DECLARE N ADDRESS
  DECLARE I BYTE
  DECLARE TOTAL ADDRESS
  ASSIGN TOTAL = 0
  ASSIGN I = 1
  DO WHILE (<= I 10)
    ASSIGN TOTAL = (+ TOTAL I)
    ASSIGN I = (+ I 1)
  CALL (CALL PUTDEC TOTAL)
```

To follow a program at run time, assemble with a listing (`lwasm ... --list=prog.lst`) and run `sim6809 -t`. The trace addresses match the listing, and the listing shows the PL/M source line for each group of instructions.

10 Testing

`make test` runs all of these suites:

Suite	What it checks
`lex`, `parse`	`--tokens` and `--ast` output for every number form, every keyword, precedence, dangling `ELSE` and nested procedures, against golden files
`err`	44 programs that must fail, each with the expected message and position
`run`	complete programs compiled, assembled and run in `sim6809`; output and exit status compared. They cover every operator and conversion rule, control flow, input, procedures (including nested calls), `HALT`, `--org` and `--vectors`
`fuzz`	random programs with thousands of random expressions, checked against an independent C model of the typing rules; `make fuzz-test N=2000` runs more
`coco`, `xroar`	the Color Computer hooks, in the simulator and on an emulated CoCo 2 (the `xroar` suite is skipped if XRoar, the ROMs or toolshed aren't installed)

Suite	What it checks
`sim`	the simulator itself (see the sim6809 manual)

11 Source code

File	Contents
`src/main.c`	command line, file handling
`src/lexer.c`	tokens, numbers, strings, comments
`src/parser.c`, `ast.c`	recursive-descent parser, syntax tree, `--ast`
`src/symtab.c`, `sema.c`	scopes, name resolution, types, call checks, recursion detection
`src/codegen.c`	assembly output
`runtime/plm_crt0.asm`, `plm_rt.asm`	startup code, 16-bit multiply and divide
`runtime/*/plm_hooks.asm`	hooks for the simulator, the CoCo, and the ACIA template